

Gusto-Grove And Dine

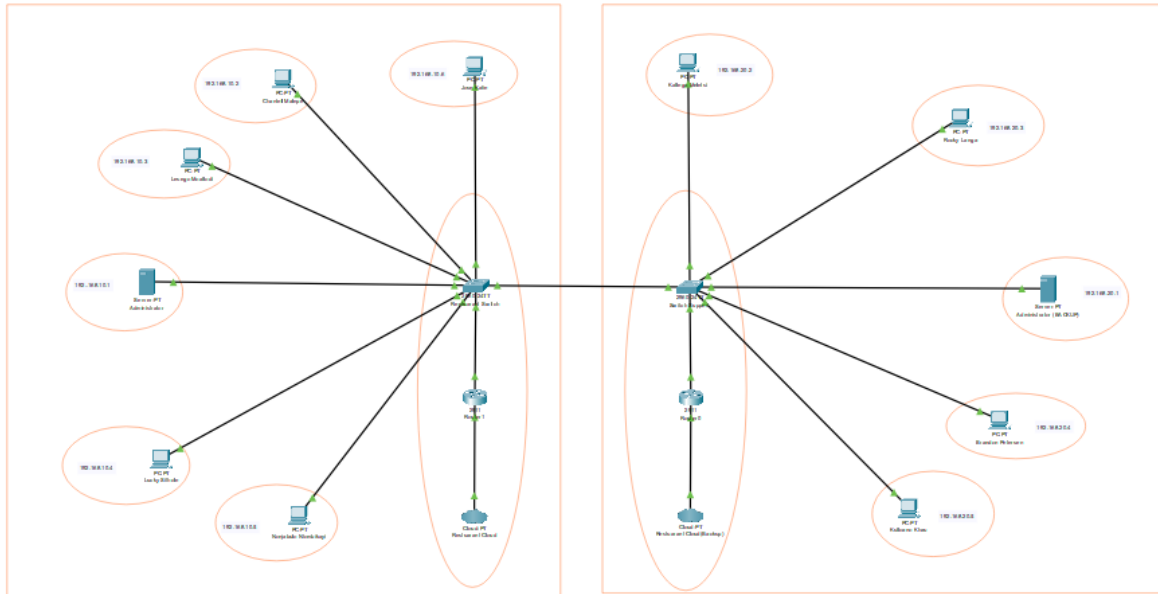
Cisco Packet Tracer (Digital Blueprint)



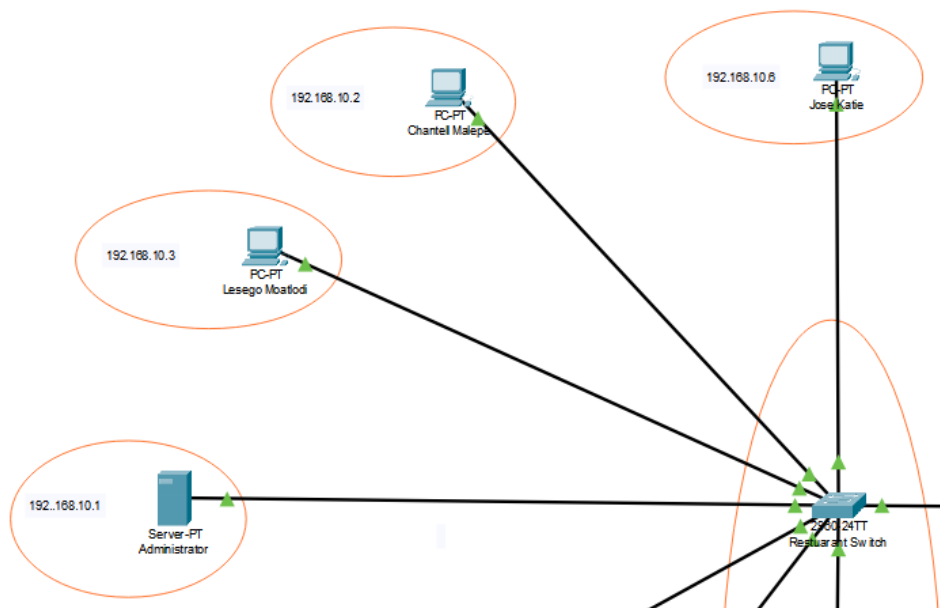


LOGICAL DESIGN

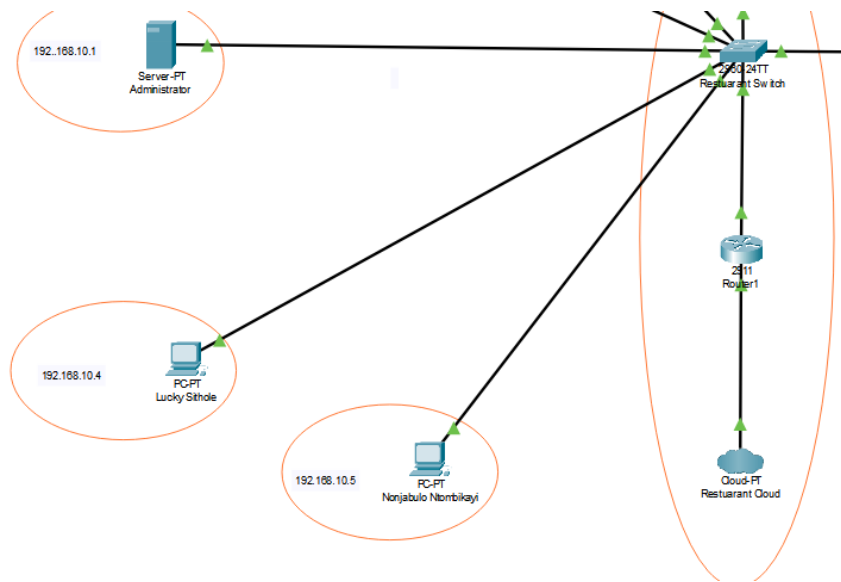
- Hybrid Topology – Bus + Star



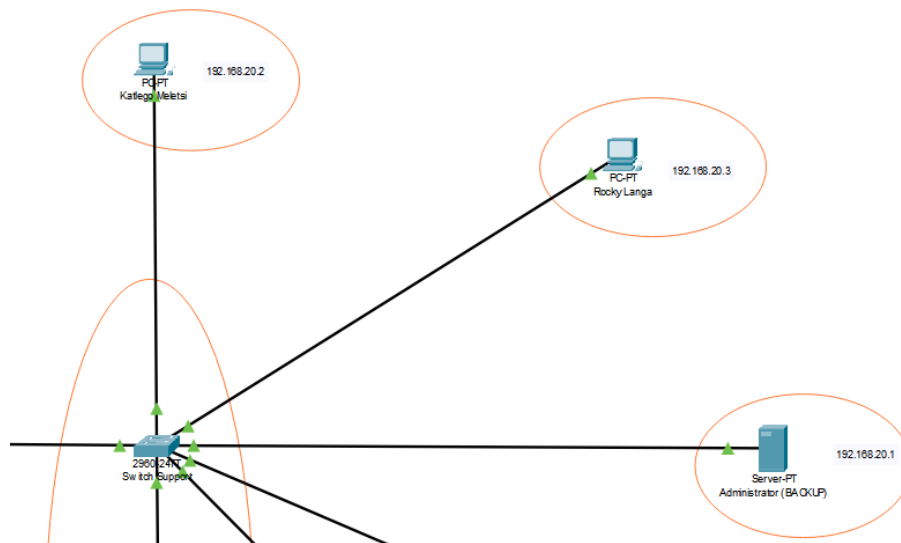
Part 1



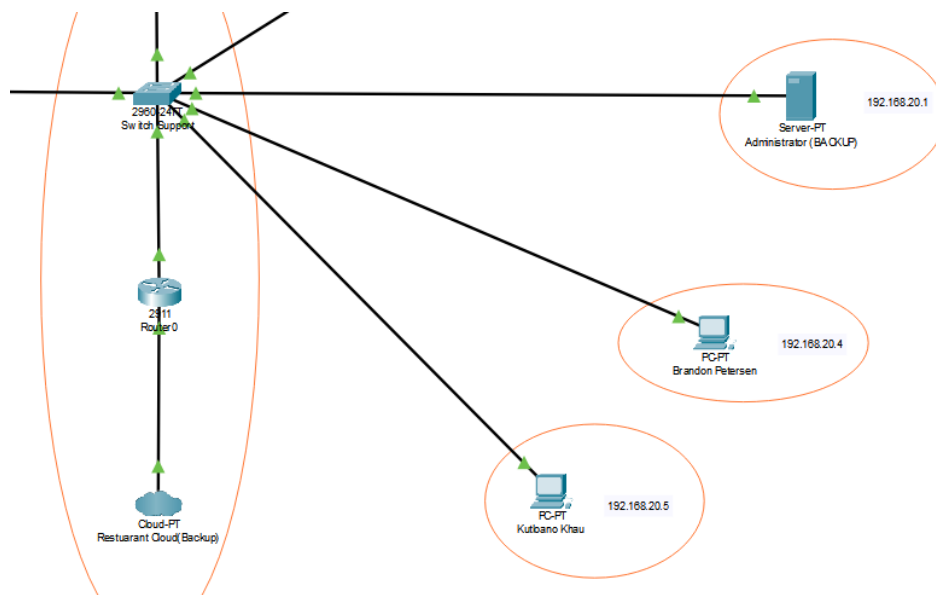
Part 2



Part 3



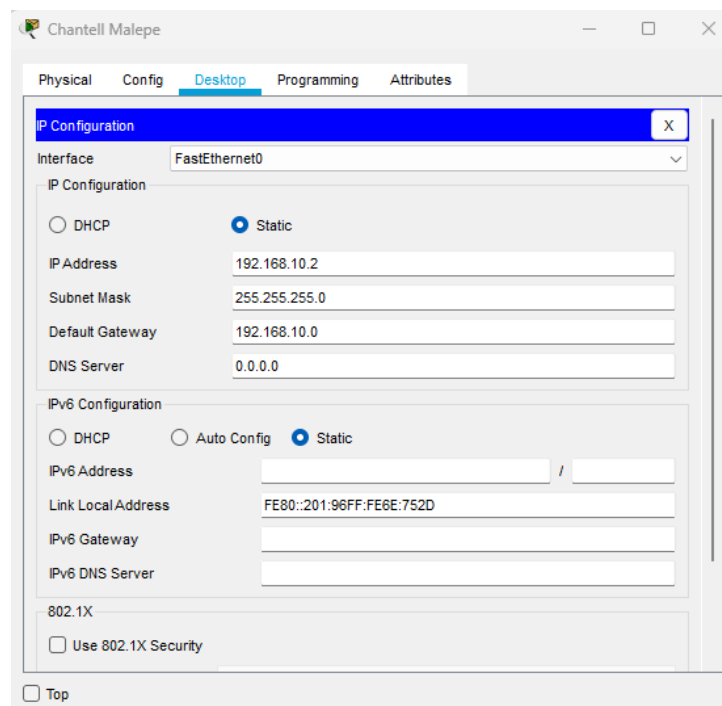
Part 4



• Configuration

- PCs

➤ Chantell Malepe



➤ Jose Katie

The screenshot shows the 'IP Configuration' window for a user named 'Jose Katie'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IP Address: 192.168.10.6, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.10.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected, with empty fields for IPv6 Address, Link Local Address (FE80::2D0:D3FF:FE3C:E26), IPv6 Gateway, and IPv6 DNS Server. A '802.1X' section at the bottom has an unchecked 'Use 802.1X Security' checkbox. A 'Top' button is at the bottom left.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	
☒ Static	
IP Address	192.168.10.6
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	
☐ Auto Config	
☒ Static	
IPv6 Address	
Link Local Address	FE80::2D0:D3FF:FE3C:E26
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	

➤ Katlego Meletsi

The screenshot shows the 'IP Configuration' window for a user named 'Katlego Meletsi'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IP Address: 192.168.20.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.20.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected, with empty fields for IPv6 Address, Link Local Address (FE80::2E0:A3FF:FE0B:8993), IPv6 Gateway, and IPv6 DNS Server. A '802.1X' section at the bottom has an unchecked 'Use 802.1X Security' checkbox. A 'Top' button is at the bottom left.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	
☒ Static	
IP Address	192.168.20.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	
☐ Auto Config	
☒ Static	
IPv6 Address	
Link Local Address	FE80::2E0:A3FF:FE0B:8993
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	

➤ Kutloano Khau

The screenshot shows the 'IP Configuration' window for the user 'Kutloano Khau'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IP Address: 192.168.20.5, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.20.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::260:47FF:FE89:6CED, IPv6 Gateway (empty), and IPv6 DNS Server (empty). At the bottom, there is a section for '802.1X' with a checkbox for 'Use 802.1X Security' which is unchecked. A 'Top' button is at the bottom left.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.20.5
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::260:47FF:FE89:6CED
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	

☐ Top

➤ Lesego Moatlodi

The screenshot shows the 'IP Configuration' window for the user 'Lesego Moatlodi'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IP Address: 192.168.10.3, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.10.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::250:FFF:FE74:730B, IPv6 Gateway (empty), and IPv6 DNS Server (empty). At the bottom, there is a section for '802.1X' with a checkbox for 'Use 802.1X Security' which is unchecked. A 'Top' button is at the bottom left.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.10.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::250:FFF:FE74:730B
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	

☐ Top

➤ Lucky Sithole

The screenshot shows the 'Lucky Sithole' application window with the 'Desktop' tab selected. The 'IP Configuration' panel is open, showing settings for the 'FastEthernet0' interface. The 'IP Configuration' section has 'Static' selected, with fields for IP Address (192.168.10.4), Subnet Mask (255.255.255.0), Default Gateway (192.168.10.0), and DNS Server (0.0.0.0). The 'IPv6 Configuration' section has 'Static' selected, with fields for IPv6 Address, Link Local Address (FE80::2D0:97FF:FE08:D12D), IPv6 Gateway, and IPv6 DNS Server. The '802.1X' section has 'Use 802.1X Security' unchecked. A 'Top' button is at the bottom left.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.10.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::2D0:97FF:FE08:D12D
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	

☐ Top

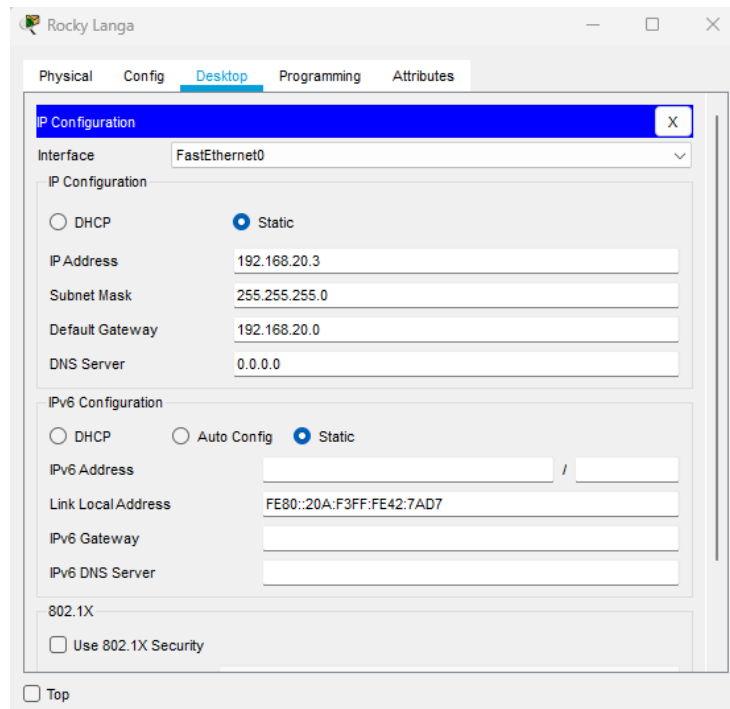
➤ Nonjabulo Ntombikayi

The screenshot shows the 'Nonjabulo Ntombikayi' application window with the 'Desktop' tab selected. The 'IP Configuration' panel is open, showing settings for the 'FastEthernet0' interface. The 'IP Configuration' section has 'Static' selected, with fields for IP Address (192.168.10.5), Subnet Mask (255.255.255.0), Default Gateway (192.168.10.0), and DNS Server (0.0.0.0). The 'IPv6 Configuration' section has 'Static' selected, with fields for IPv6 Address, Link Local Address (FE80::203:E4FF:FE15:8EAC), IPv6 Gateway, and IPv6 DNS Server. The '802.1X' section has 'Use 802.1X Security' unchecked. A 'Top' button is at the bottom left.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.10.5
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::203:E4FF:FE15:8EAC
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	

☐ Top

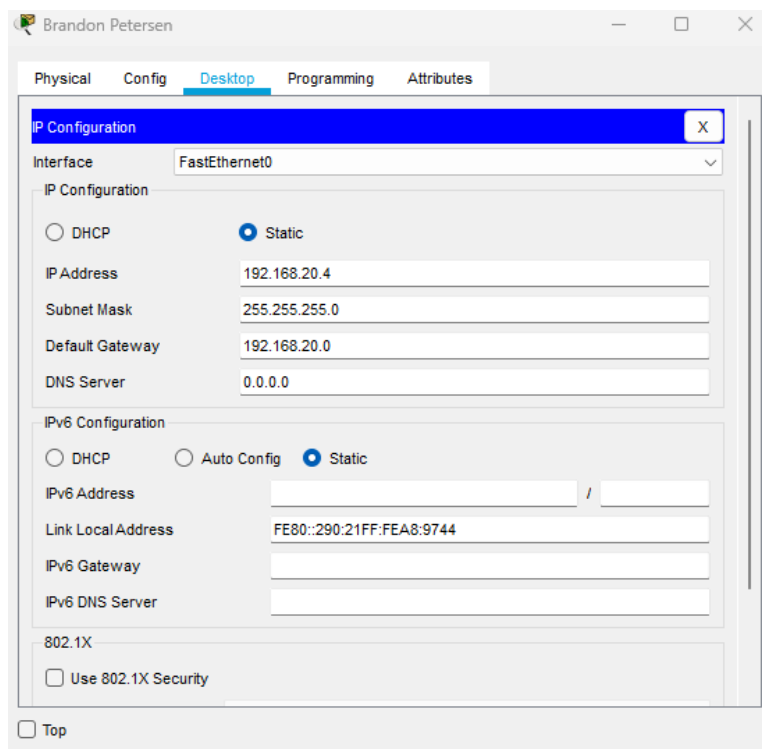
➤ Rocky Langa



The screenshot shows the 'IP Configuration' window for the user 'Rocky Langa'. The window has tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IP Address: 192.168.20.3, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.20.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::20A:F3FF:FE42:7AD7, IPv6 Gateway (empty), and IPv6 DNS Server (empty). At the bottom, there is a checkbox for 'Use 802.1X Security' which is unchecked, and a 'Top' button.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.20.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::20A:F3FF:FE42:7AD7
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	
☐ Top	

➤ Brandon Petersen



The screenshot shows the 'IP Configuration' window for the user 'Brandon Petersen'. The window has tabs for 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IP Address: 192.168.20.4, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.20.0, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::290:21FF:FEA8:9744, IPv6 Gateway (empty), and IPv6 DNS Server (empty). At the bottom, there is a checkbox for 'Use 802.1X Security' which is unchecked, and a 'Top' button.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.20.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	
Link Local Address	FE80::290:21FF:FEA8:9744
IPv6 Gateway	
IPv6 DNS Server	
802.1X	
☐ Use 802.1X Security	
☐ Top	

- Routers

➤ Admin Router

The screenshot shows the 'Admin Router' configuration window. The 'Config' tab is active, and the 'SWITCHING' section is expanded, showing the 'VLAN Database' sub-section. The 'VLAN Configuration' table is visible, showing a single entry for VLAN 10 with the name 'admin'. The 'Equivalent IOS Commands' section at the bottom displays the commands used to configure the VLAN.

VLAN No	VLAN Name
1	default
10	admin
1002	fdi-default
1003	token-ring-default
1004	fdinet-default
1005	trnet-default

```
APPLY completed.
Exiting...
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#end
Router#vlan database
Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.
Router(vlan)#
VLAN 10 modified:
  Name: admin
Router(vlan)#
```

➤ Backup Router

The screenshot shows the 'Backup Router' configuration window. The 'Config' tab is active, and the 'SWITCHING' section is expanded, showing the 'VLAN Database' sub-section. The 'VLAN Configuration' table is visible, showing a single entry for VLAN 20 with the name 'backup'. The 'Equivalent IOS Commands' section at the bottom displays the commands used to configure the VLAN.

VLAN No	VLAN Name
1	default
20	backup
1002	fdi-default
1003	token-ring-default
1004	fdinet-default
1005	trnet-default

```
$LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
$LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#enable
Router#
Router#vlan database
Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.
Router(vlan)#vlan 20 name backup
VLAN 20 modified:
  Name: backup
Router(vlan)#
```

- Server

➤ Administrator Server

The screenshot shows the 'Administrator' window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the following settings:

Configuration Type	Value
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.0
DNS Server	0.0.0.0

The 'IPv6 Configuration' section is also expanded, showing the following settings:

Configuration Type	Value	
☐ DHCP	☐ Auto Config	☒ Static
IPv6 Address		
Link Local Address	FE80::260:47FF:FE04:51C4	
IPv6 Gateway		
IPv6 DNS Server		

The '802.1X' section is expanded, showing the following settings:

Configuration Type	Value
☐ Use 802.1X Security	
Authentication	MD5

At the bottom of the window, there is a 'Top' button.

➤ Backup Server

The screenshot shows the 'Administrator (BACKUP)' window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing the following settings:

Configuration Type	Value
IP Configuration	
☐ DHCP	☒ Static
IP Address	192.168.20.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.0
DNS Server	0.0.0.0

The 'IPv6 Configuration' section is also expanded, showing the following settings:

Configuration Type	Value	
☐ DHCP	☐ Auto Config	☒ Static
IPv6 Address		
Link Local Address	FE80::260:3EFF:FEC2:3738	
IPv6 Gateway		
IPv6 DNS Server		

The '802.1X' section is expanded, showing the following settings:

Configuration Type	Value
☐ Use 802.1X Security	
Authentication	MD5

At the bottom of the window, there is a 'Top' button.

- Switches

➤ Restaurant Switch

The screenshot shows the 'Restaurant Switch' configuration window. The 'Config' tab is active. On the left, the 'VLAN Database' is selected under the 'SWITCHING' section. The main area displays the 'VLAN Configuration' section with fields for 'VLAN Number' (10) and 'VLAN Name' (admin). Below these are 'Add' and 'Remove' buttons. A table lists existing VLANs: 1 (default), 10 (admin), 1002 (fddi-default), 1003 (token-ring-default), 1004 (fddinet-default), and 1005 (trnet-default). At the bottom, a warning message states: 'Warning: It is recommended to configure VLAN from config mode, as VLAN database mode is being deprecated. Please consult user documentation for configuring VTP/VLAN in config mode.' Below the warning is a text box containing 'Switch(vlan)#'.

Restuarant Switch

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

VLAN Configuration

VLAN Number: 10

VLAN Name: admin

Add Remove

VLAN No	VLAN Name
1	default
10	admin
1002	fddi-default
1003	token-ring-default
1004	fddinet-default
1005	trnet-default

Equivalent IOS Commands

Warning: It is recommended to configure VLAN from config mode, as VLAN database mode is being deprecated. Please consult user documentation for configuring VTP/VLAN in config mode.

Switch(vlan)#

☐ Top

➤ Switch Support

The screenshot shows the 'Switch Support' configuration window. The 'Config' tab is active. On the left, the 'VLAN Database' is selected under the 'SWITCHING' section. The main area displays the 'VLAN Configuration' section with fields for 'VLAN Number' (20) and 'VLAN Name' (backup). Below these are 'Add' and 'Remove' buttons. A table lists existing VLANs: 1 (default), 20 (backup), 1002 (fddi-default), 1003 (token-ring-default), 1004 (fddinet-default), and 1005 (trnet-default). At the bottom, a warning message states: 'Warning: It is recommended to configure VLAN from config mode, as VLAN database mode is being deprecated. Please consult user documentation for configuring VTP/VLAN in config mode.' Below the warning is a text box containing 'Switch(vlan)#'.

Switch Support

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

VLAN Configuration

VLAN Number: 20

VLAN Name: backup

Add Remove

VLAN No	VLAN Name
1	default
20	backup
1002	fddi-default
1003	token-ring-default
1004	fddinet-default
1005	trnet-default

Equivalent IOS Commands

Warning: It is recommended to configure VLAN from config mode, as VLAN database mode is being deprecated. Please consult user documentation for configuring VTP/VLAN in config mode.

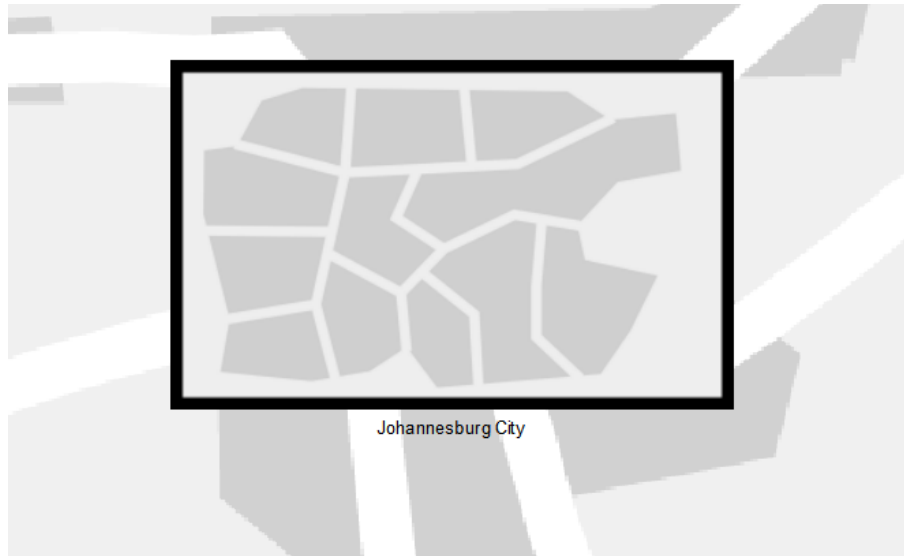
Switch(vlan)#

☐ Top

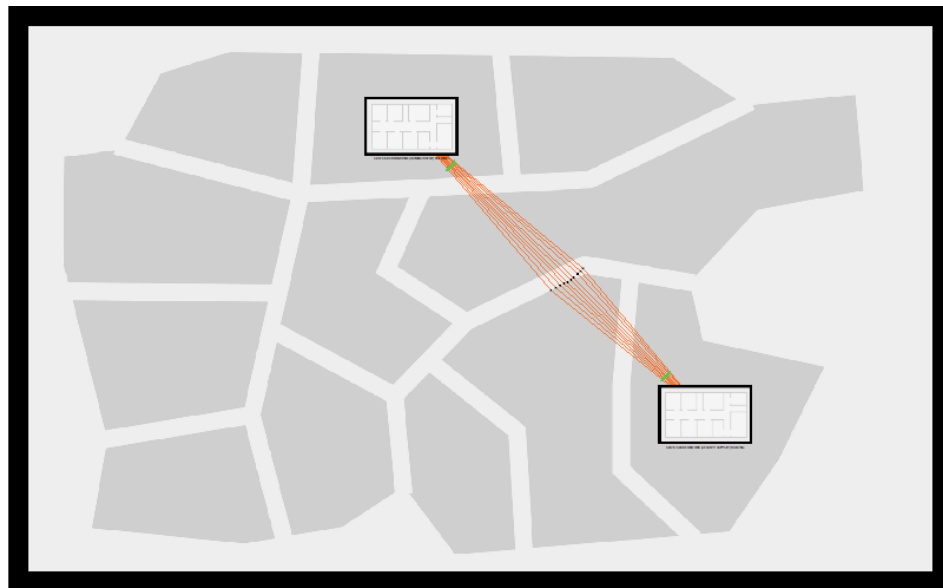


PHYSICAL DESIGN

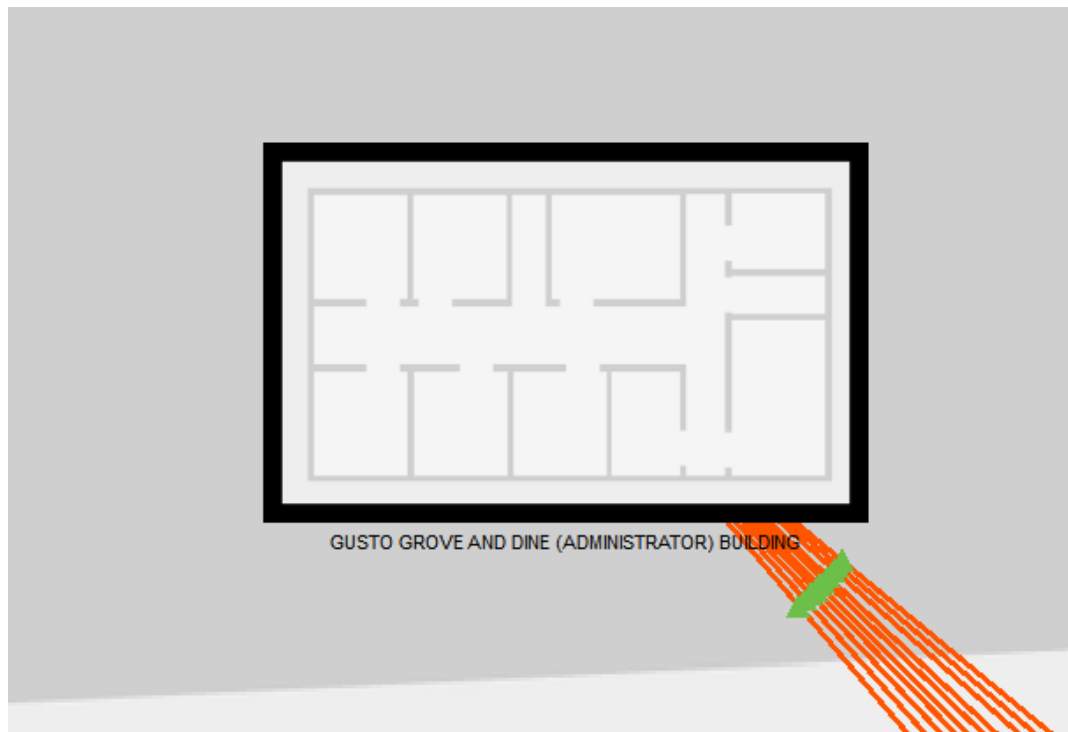
- New City
 - Johannesburg



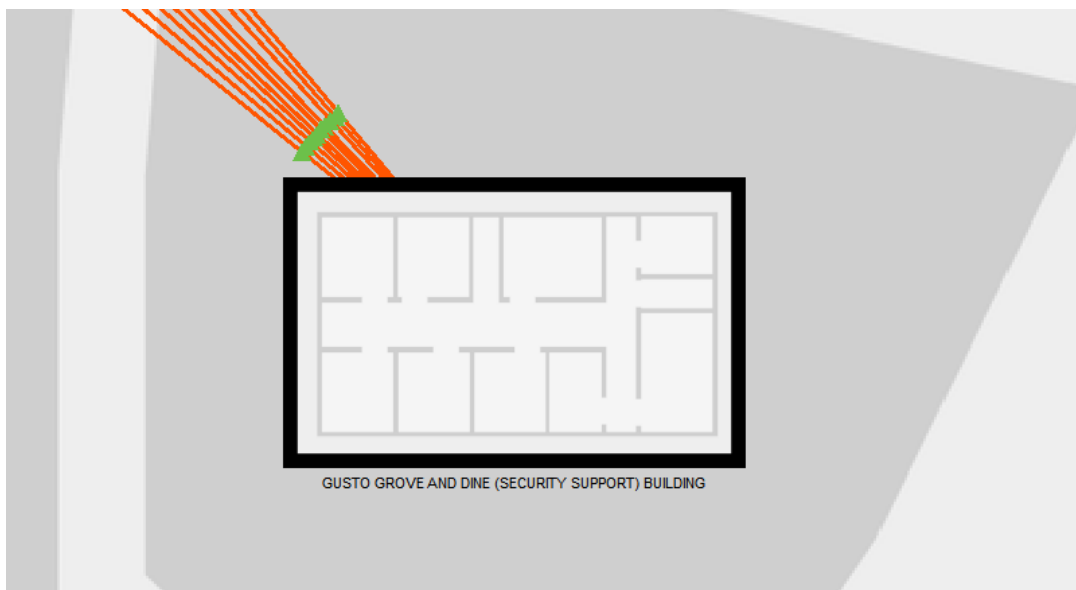
Part 1



Part 2

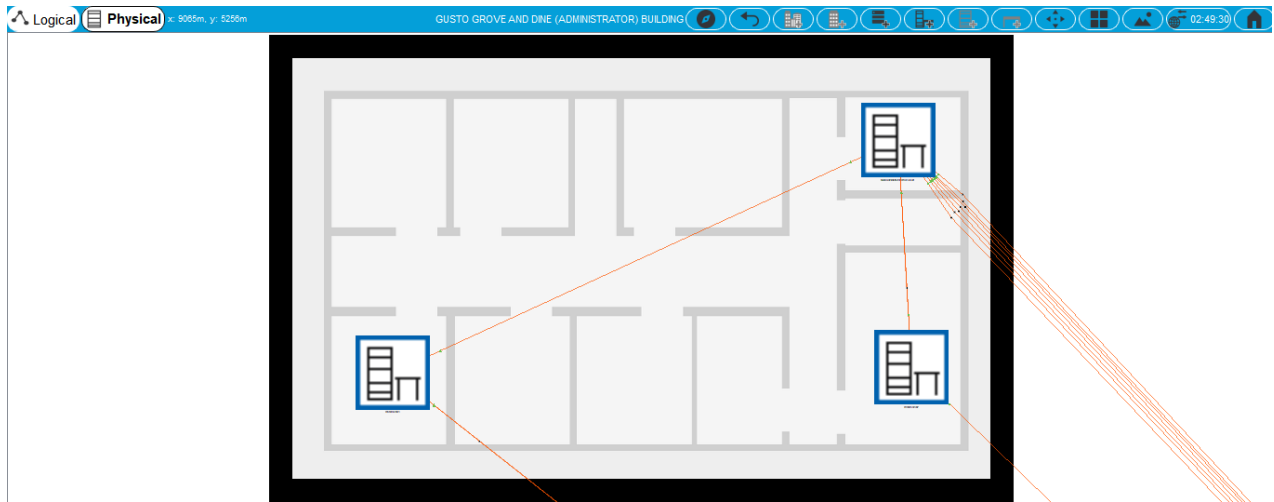


Part 3

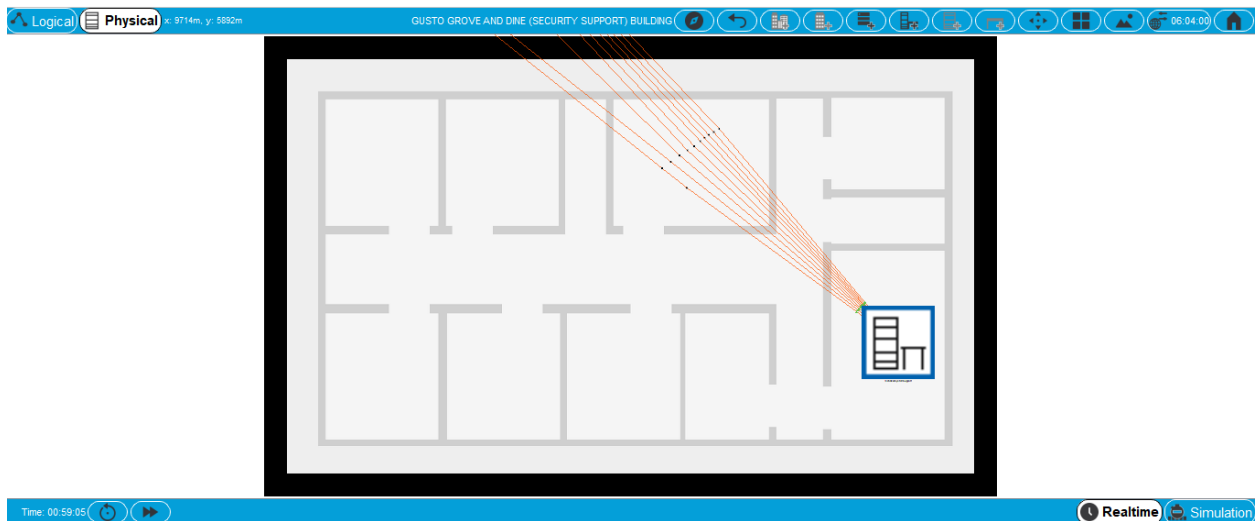


- Office Buildings

GUSTO-GROVE AND DINE (ADMINISTRATOR) BUILDING



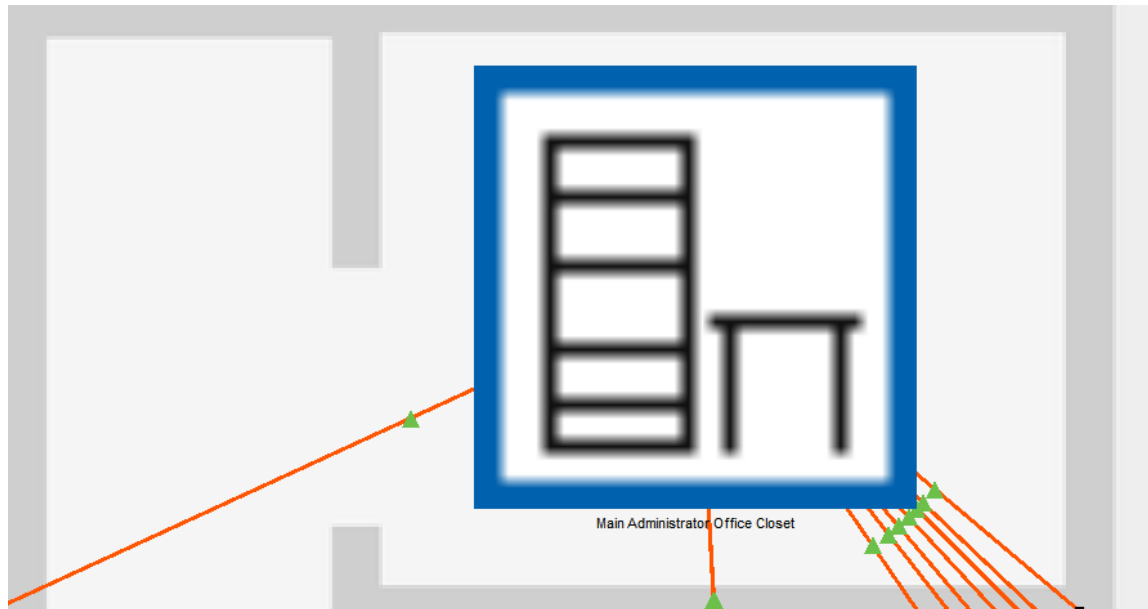
GUSTO-GROVE AND DINE (SECURITY SUPPORT) BUILDING



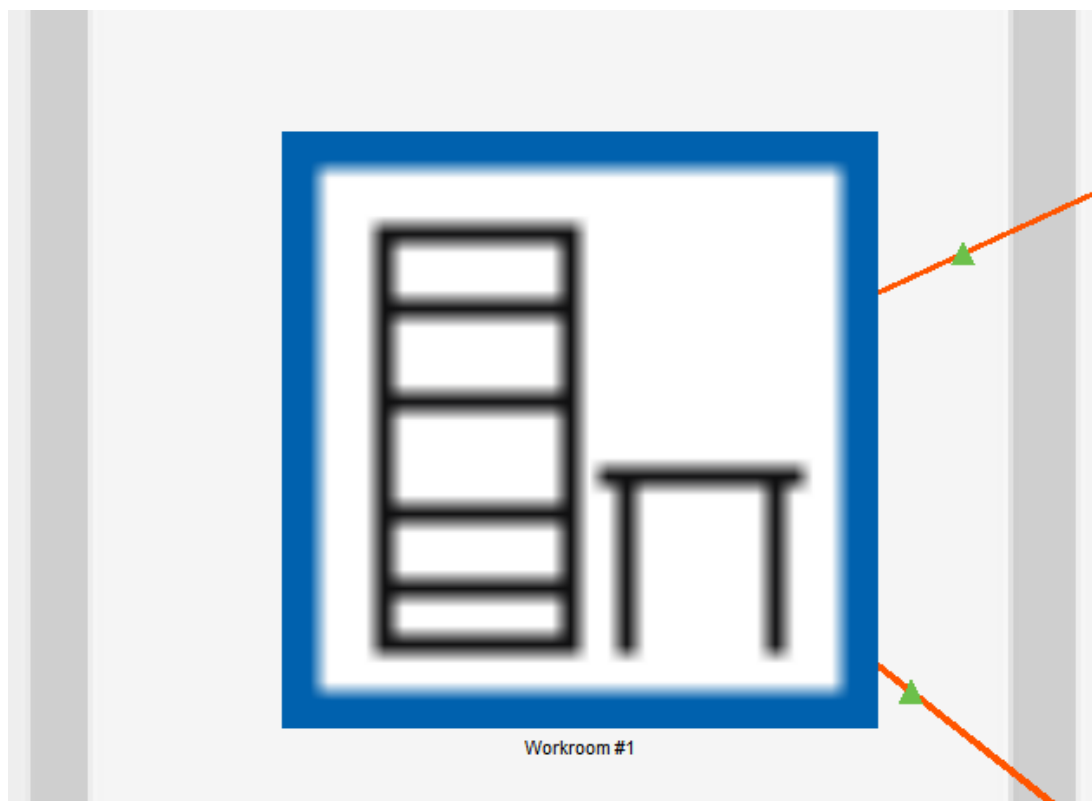
CLOSET SPACES

➤ ADMIN

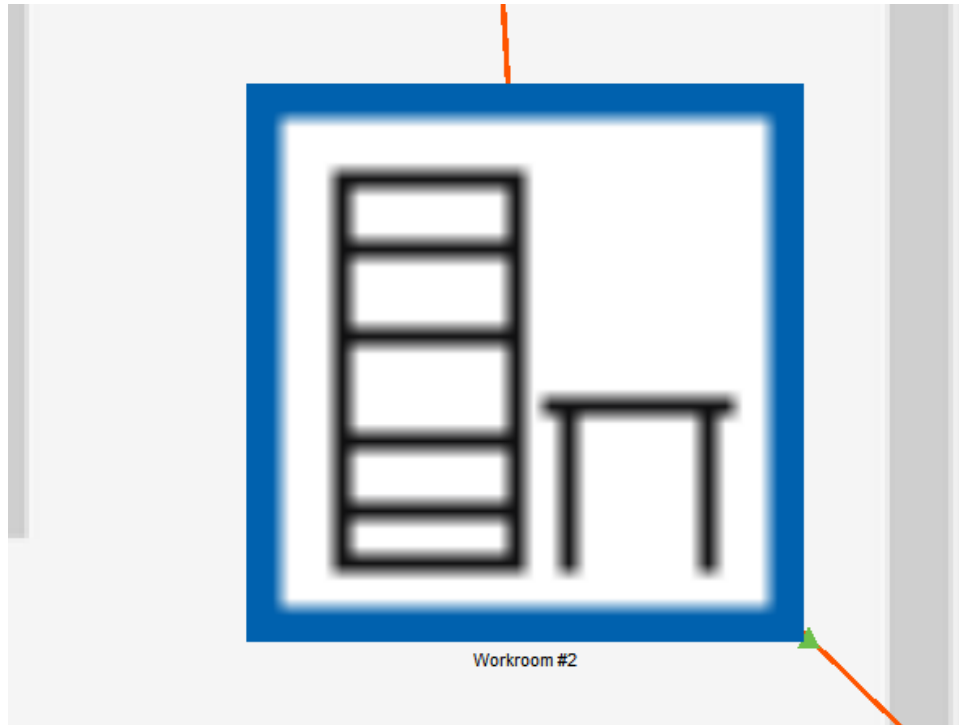
Part 1



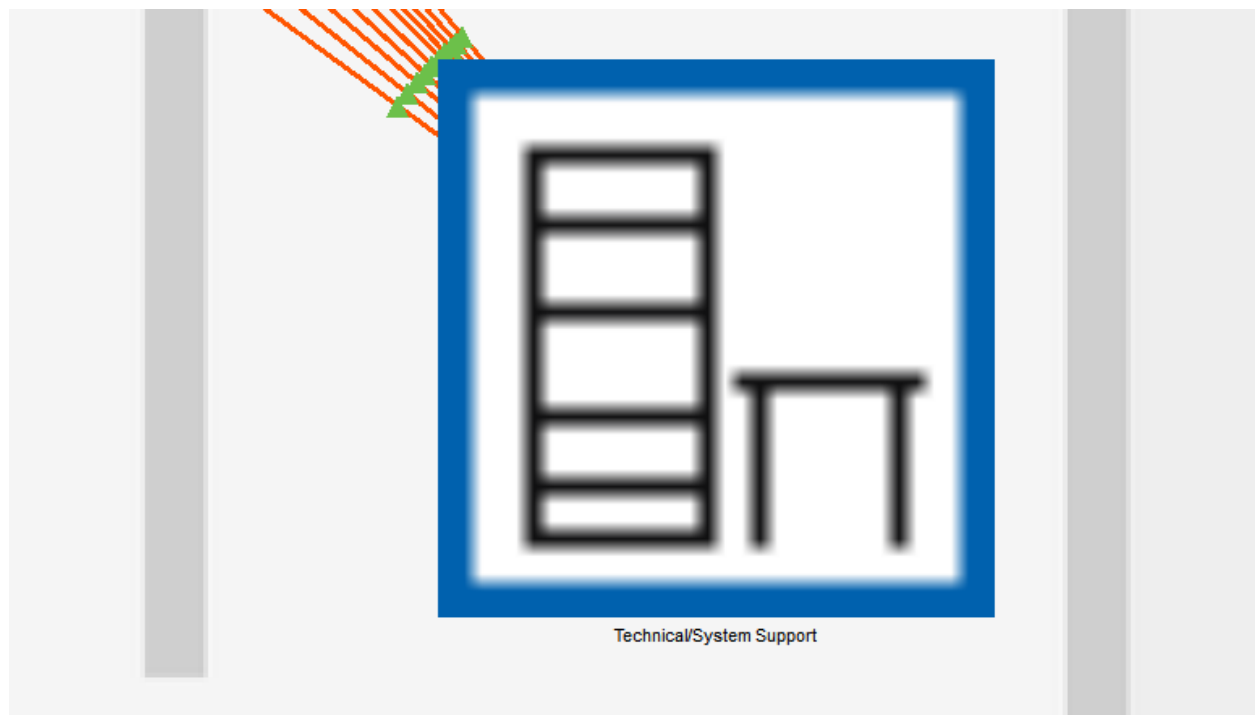
Part 2



Part 3



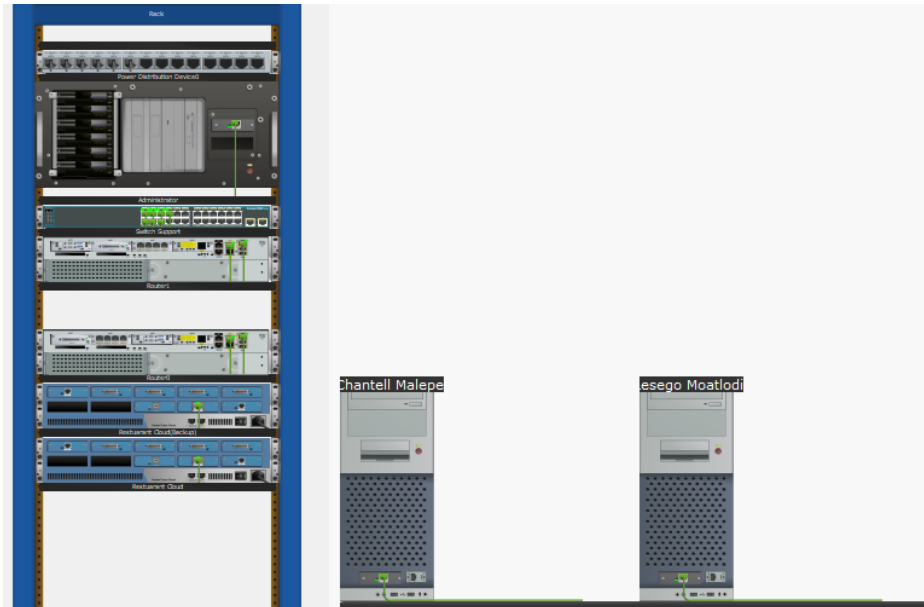
➤ BACKUP



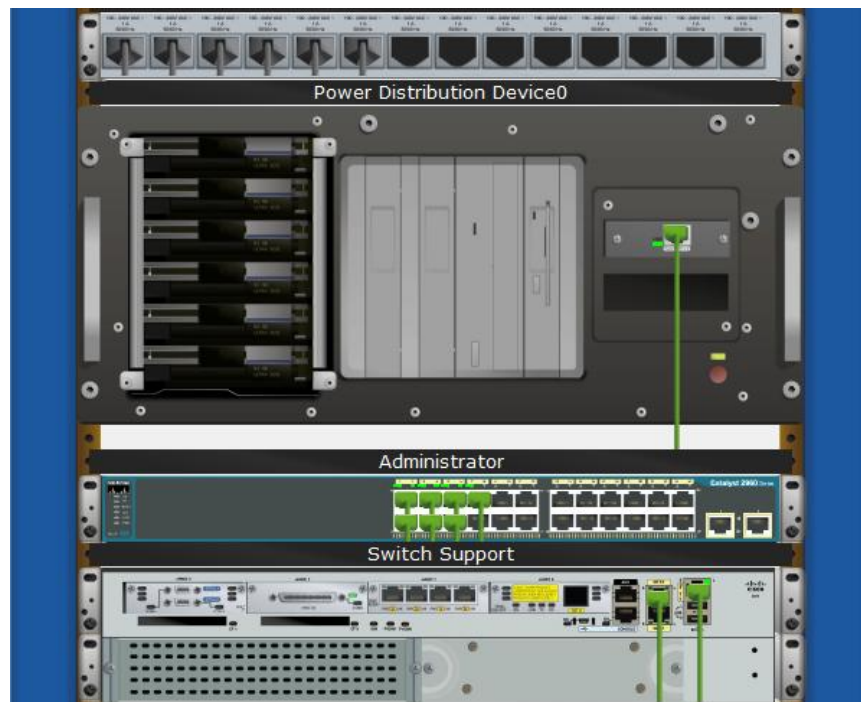
• HARDWARE

- ADMIN

➤ MAIN ADMINISTRATOR CLOSET STORAGE



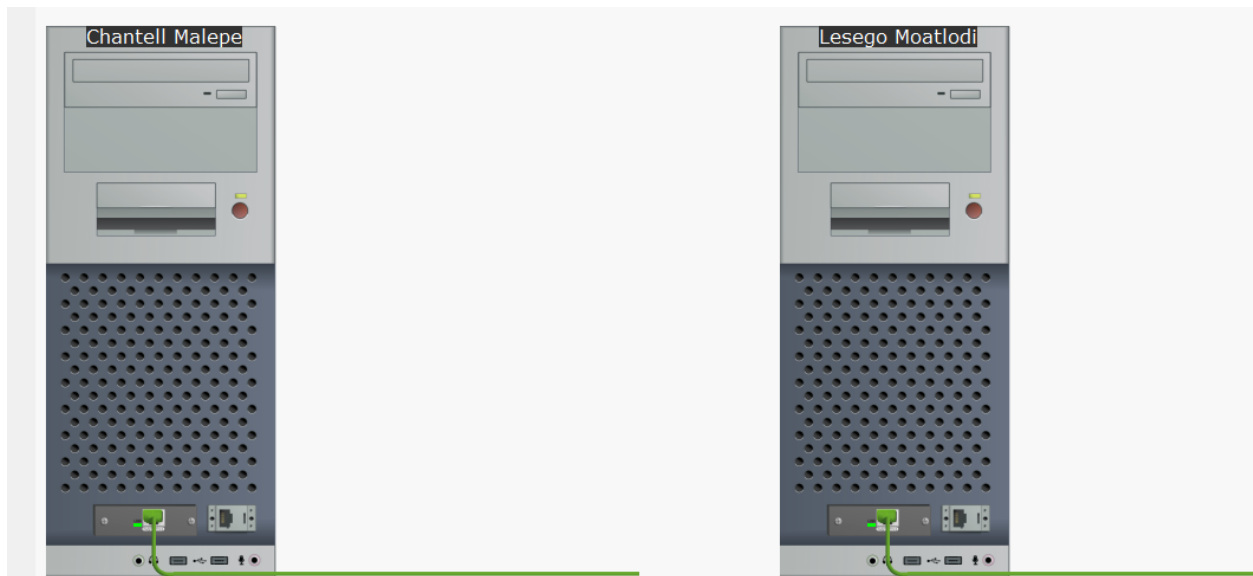
➤ MAIN ADMINISTRATOR CLOSET STORAGE PART 1



➤ MAIN ADMINISTRATOR CLOSET STORAGE PART 2

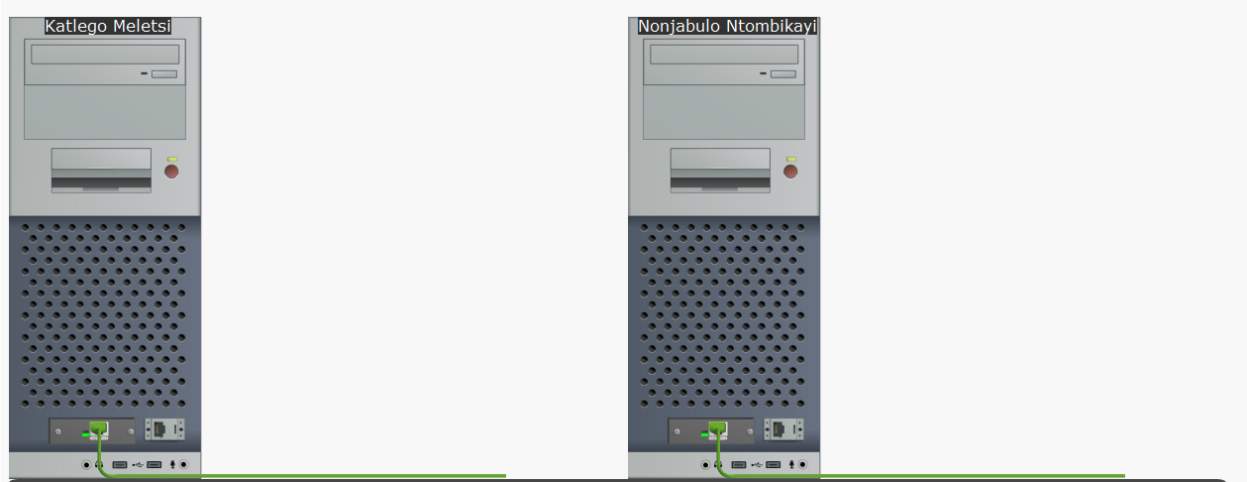


➤ MAIN ADMINISTRATOR CLOSET STORAGE PCs

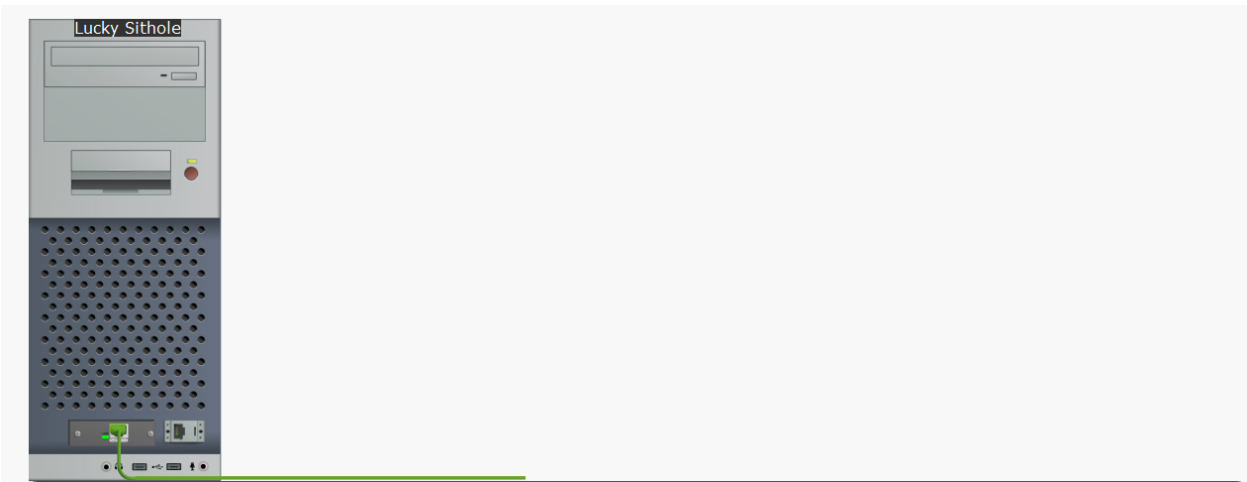


➤ WORKSTATION #1

PART 1



PART 2

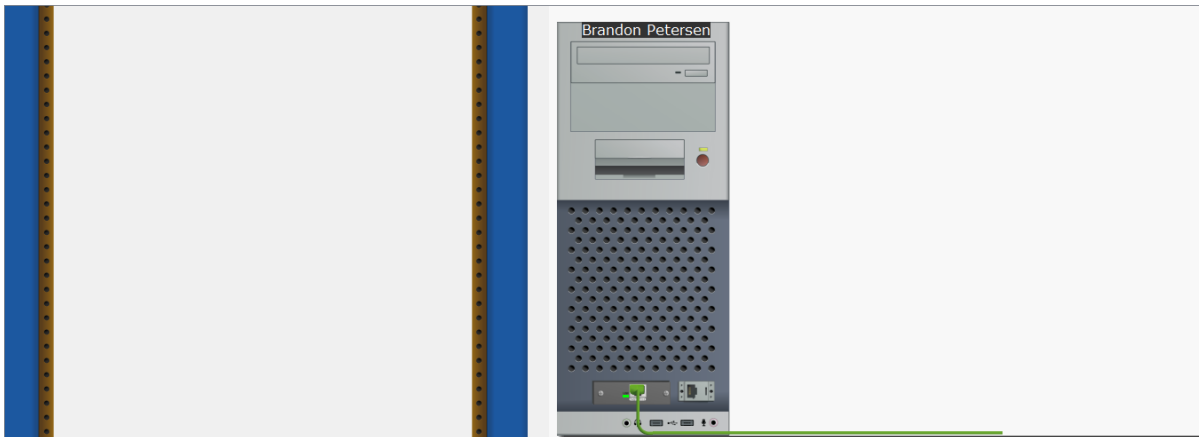


➤ WORKSTATION #2



➤ TECHNICAL/SYSTEM SUPPORT

PC SUPPORT

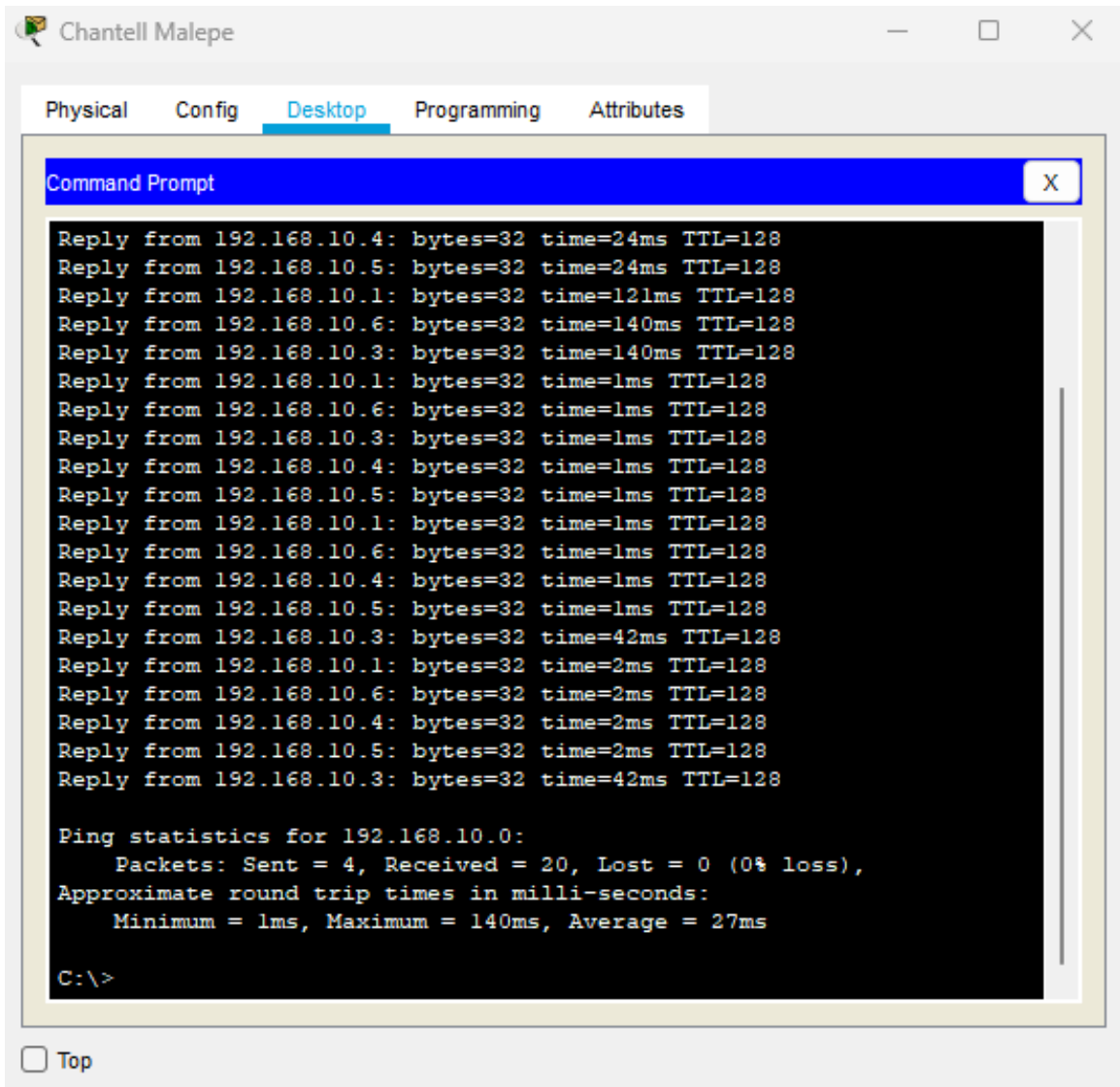


RACK SUPPORT



❖ TESTING PHASE

- PING COMMAND PROMPT



The screenshot shows a Windows desktop environment with a window titled "Chantell Malepe". Inside the window, there are tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, and a "Command Prompt" window is open. The Command Prompt displays the output of a ping command to 192.168.10.0. The output shows 20 successful replies from various IP addresses in the 192.168.10.0/24 range, with varying response times and TTL values. Below the replies, the ping statistics are shown, indicating 4 packets sent, 20 received, and 0% loss. The approximate round trip times are: Minimum = 1ms, Maximum = 140ms, Average = 27ms. The Command Prompt prompt is "C:\>".

```
Reply from 192.168.10.4: bytes=32 time=24ms TTL=128
Reply from 192.168.10.5: bytes=32 time=24ms TTL=128
Reply from 192.168.10.1: bytes=32 time=121ms TTL=128
Reply from 192.168.10.6: bytes=32 time=140ms TTL=128
Reply from 192.168.10.3: bytes=32 time=140ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=42ms TTL=128
Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Reply from 192.168.10.6: bytes=32 time=2ms TTL=128
Reply from 192.168.10.4: bytes=32 time=2ms TTL=128
Reply from 192.168.10.5: bytes=32 time=2ms TTL=128
Reply from 192.168.10.3: bytes=32 time=42ms TTL=128

Ping statistics for 192.168.10.0:
    Packets: Sent = 4, Received = 20, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 140ms, Average = 27ms

C:\>
```